

BY SIR OTONDE MICHAEL -DELEGATE

KRM 300 SAMPLED PAST PAPERS WITH ANSWERS

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1. Key characteristics of a fertile soil (5 marks)

1. **Adequate nutrient supply** – Contains essential macro- and micro-nutrients in available forms for plant uptake.
2. **Good soil structure** – Well-aggregated particles that allow proper root penetration and aeration.
3. **Proper soil pH** – Slightly acidic to neutral (about 6.0–7.5) for optimum nutrient availability.
4. **High organic matter content** – Improves nutrient supply, water retention, and microbial activity.
5. **Good water-holding capacity and drainage** – Retains sufficient moisture without becoming waterlogged.

2. Four pathways through which Nitrogen is lost from the soil system (4 marks)

1. **Leaching** – Nitrates are washed down beyond the root zone by rainfall or irrigation.
2. **Volatilization** – Loss of ammonia gas to the atmosphere, especially from surface-applied urea.
3. **Denitrification** – Conversion of nitrates into nitrogen gases under waterlogged conditions.
4. **Crop removal** – Nitrogen is taken up by plants and removed during harvesting.

3. Difference between complete and incomplete fertilizers (5 marks)

1. **Complete fertilizers** contain all three primary nutrients: Nitrogen (N), Phosphorus (P), and Potassium (K).
2. **Incomplete fertilizers** contain only one or two of the primary nutrients.
3. Complete fertilizers are suitable where soils lack multiple nutrients.
4. Incomplete fertilizers are used to correct specific nutrient deficiencies.
5. Example: NPK 17:17:17 is complete; Urea (46% N) is incomplete.

4. Difference between macro-nutrients and micro-nutrients in soils (5 marks)

1. **Macro-nutrients** are required in large quantities by plants.
2. **Micro-nutrients** are required in small (trace) amounts.
3. Macro-nutrients include Nitrogen (N), Phosphorus (P), Potassium (K), Calcium (Ca), Magnesium (Mg), Sulfur (S).
4. Micro-nutrients include Iron (Fe), Zinc (Zn), Copper (Cu), Manganese (Mn), Boron (B), Molybdenum (Mo).

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5. Deficiency of either type affects plant growth, but macro-nutrient deficiencies usually show more rapid and widespread symptoms.

5. Example of nitrogenous fertilizers and how it should be applied (5 marks)

Example: Urea (46% N)

1. Apply at recommended rates based on crop requirements.
2. Incorporate into the soil to reduce nitrogen loss through volatilization.
3. Apply in split doses (e.g., at planting and during growth stages).
4. Apply when soil moisture is adequate for better absorption.
5. Avoid application before heavy rains to reduce leaching losses.

6. Two physiological functions of phosphorus in plants (5 marks)

1. **Energy transfer** – Essential component of ATP used in energy storage and transfer.
2. **Root development** – Promotes strong root growth and early plant establishment.
3. **Cell division** – Important in DNA and RNA formation.
4. **Flowering and seed formation** – Encourages reproductive growth.
5. **Maturity** – Enhances early crop maturity.

7. Two ways in which soil pH affects nutrient availability (5 marks)

1. **Influences solubility of nutrients** – Some nutrients become unavailable in very acidic or alkaline soils.
2. **Affects microbial activity** – Microorganisms responsible for nutrient cycling function best at optimal pH.
3. At low pH, aluminum and manganese may become toxic.
4. At high pH, phosphorus may become fixed and unavailable.
5. Certain nutrients like iron become less available in alkaline soils.

8. Role of liming in soil fertility management (5 marks)

1. **Raises soil pH** – Reduces soil acidity.
2. **Improves nutrient availability** – Makes phosphorus and other nutrients more available.
3. **Reduces toxicity** – Neutralizes toxic aluminum and manganese in acidic soils.
4. **Enhances microbial activity** – Improves decomposition and nitrogen fixation.

9. Role of soil organic matter in soil physical, chemical, and biological processes (15 marks)

A. Physical roles

1. **Improves soil structure** – Promotes aggregation of soil particles into stable crumbs.
2. **Enhances soil porosity** – Increases pore spaces for air and water movement.

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3. **Increases water-holding capacity** – Organic matter absorbs and retains moisture.
4. **Improves infiltration rate** – Reduces surface runoff and erosion.
5. **Reduces soil compaction** – Makes soil easier for root penetration.

B. Chemical roles

6. **Acts as a nutrient reservoir** – Stores essential nutrients like N, P, and S.
7. **Improves cation exchange capacity (CEC)** – Enhances the soil's ability to hold nutrients.
8. **Buffers soil pH** – Reduces rapid changes in soil acidity or alkalinity.
9. **Chelates micronutrients** – Keeps elements like Fe and Zn available to plants.
10. **Reduces nutrient leaching** – Holds nutrients in the root zone.

C. Biological roles

11. **Provides energy for microorganisms** – Supplies carbon as a food source.
12. **Enhances microbial activity** – Stimulates decomposition and nutrient cycling.
13. **Supports nitrogen fixation** – Encourages beneficial bacteria.
14. **Promotes earthworm activity** – Improves soil aeration and mixing.
15. **Suppresses some soil-borne diseases** – Encourages beneficial organisms that outcompete pathogens.

10. Factors limiting biological nitrogen fixation in smallholder farms (15 marks)

1. **Low soil fertility** – Deficiency of phosphorus and other nutrients reduces fixation.
2. **Soil acidity (low pH)** – Inhibits growth of nitrogen-fixing bacteria.
3. **High soil temperature** – Reduces survival of rhizobia bacteria.
4. **Drought stress** – Limits bacterial activity and root growth.
5. **Waterlogging** – Reduces oxygen needed by bacteria.
6. **Poor quality legume seeds** – Weak plants form fewer nodules.
7. **Absence of effective rhizobia strains** – Soil may lack compatible bacteria.
8. **Lack of inoculation** – Farmers may not use rhizobial inoculants.
9. **Continuous monocropping** – Reduces soil microbial diversity.
10. **Pest and disease attack** – Damages legumes and reduces nodulation.
11. **Excess nitrogen fertilizer** – Suppresses biological nitrogen fixation.
12. **Low organic matter** – Reduces microbial activity.
13. **Poor agronomic practices** – Late planting or improper spacing.
14. **Salinity problems** – Salt stress inhibits bacterial activity.
15. **Limited farmer knowledge and resources** – Lack of awareness about inoculation and soil management.

11. Approaches to increase phosphorus availability in tropical soils (15 marks)

1. **Application of phosphorus fertilizers** – Use of DAP, TSP, or SSP.
2. **Band placement of P fertilizers** – Reduces fixation in tropical soils.

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3. **Use of rock phosphate in acidic soils** – Slowly releases phosphorus.
4. **Liming acidic soils** – Raises pH and reduces P fixation by aluminum and iron.
5. **Incorporation of organic matter** – Organic acids reduce phosphorus fixation.
6. **Use of compost and manure** – Supplies phosphorus and improves availability.
7. **Crop rotation with legumes** – Enhances soil fertility and P cycling.
8. **Use of phosphate-solubilizing microorganisms** – Improves P release.
9. **Intercropping systems** – Some crops enhance phosphorus uptake.
10. **Use of mycorrhizal fungi** – Improves phosphorus absorption by roots.
11. **Conservation tillage** – Maintains organic matter and reduces P losses.
12. **Balanced fertilization** – Ensures other nutrients do not limit P uptake.
13. **Use of acid-tolerant crop varieties** – Better adapted to low P conditions.
14. **Proper timing of application** – Apply at planting for early root development.
15. **Soil testing and site-specific management** – Apply appropriate P rates to avoid deficiency or wastage.

1. Explain two indicators of fertility status of soil in a farmer's field (5 marks)

1. **Crop performance and vigor** – Healthy, dark green crops with good growth indicate fertile soil, while stunted growth suggests nutrient deficiency.
2. **Yield levels** – High and consistent yields over seasons indicate good fertility; declining yields show nutrient depletion.
3. **Presence of deficiency symptoms** – Yellowing (N deficiency), purpling (P deficiency), or scorched leaf margins (K deficiency) indicate low fertility.
4. **Soil colour** – Dark soils usually indicate high organic matter and better fertility; pale soils suggest low organic matter.
5. **Soil testing results** – Laboratory analysis showing adequate nutrient levels and suitable pH confirms good fertility status.

2. Differentiate between complete and incomplete fertilizers (5 marks)

1. **Complete fertilizers** contain all three primary nutrients (N, P, K).
2. **Incomplete fertilizers** contain only one or two of the primary nutrients.
3. Complete fertilizers are used where multiple nutrient deficiencies exist.
4. Incomplete fertilizers are used to correct specific nutrient deficiencies.
5. Example: NPK 20:20:20 (complete); Urea (46% N) or TSP (P only) (incomplete).

3. Explain the disadvantages of blended fertilizers (5 marks)

1. **Nutrient segregation** – Nutrients may separate during transport, leading to uneven application.
2. **Non-uniform nutrient distribution** – Crops may receive unequal proportions of nutrients.
3. **Limited flexibility after blending** – Nutrient ratios cannot easily be adjusted once mixed.
4. **Possible incompatibility** – Some fertilizer materials react chemically when blended.

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5. **Quality control issues** – Poor blending may lead to incorrect nutrient composition.

4. Explain the role of phosphorus in plants (5 marks)

1. **Energy transfer** – Component of ATP for energy storage and transfer.
2. **Root development** – Promotes strong root growth and early establishment.
3. **Cell division and growth** – Essential in DNA and RNA formation.
4. **Flowering and seed formation** – Enhances reproductive development.
5. **Early maturity** – Speeds up crop maturity and improves yield quality.

5. State three harmful effects on plants resulting from excessive supply of nitrogen (6 marks)

1. **Excessive vegetative growth** – Leads to weak stems and lodging.
2. **Delayed maturity** – Crops take longer to flower and mature.
3. **Reduced fruiting and grain filling** – More leaves than reproductive growth.
4. **Increased pest and disease susceptibility** – Tender growth attracts pests.
5. **Nutrient imbalance** – Interferes with uptake of other nutrients like potassium.
6. **Poor storage quality** – Produce may have lower keeping quality.

6. Briefly explain ways through which organic fertilizers contribute to plant nutrient availability (5 marks)

1. **Mineralization** – Microorganisms decompose organic matter and release nutrients.
2. **Improved soil structure** – Enhances root growth and nutrient uptake.
3. **Increased CEC** – Improves soil's ability to retain nutrients.
4. **Reduced nutrient leaching** – Organic matter holds nutrients in the root zone.
5. **Stimulation of soil microbes** – Enhances nutrient cycling and availability.

7. Give two examples of nitrogenous fertilizers and state how they should be applied for optimum nutrient utilization (5 marks)

Examples:

- Urea (46% N)
 - Ammonium sulphate
1. Apply at recommended rates based on crop requirement.
 2. Incorporate into soil to reduce volatilization losses (especially urea).
 3. Apply in split doses during crop growth stages.
 4. Apply when soil moisture is adequate.
 5. Avoid application before heavy rainfall to reduce leaching.

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8. Explain the role of liming in soil fertility management (5 marks)

1. Raises soil pH by reducing acidity.
2. Reduces aluminum and manganese toxicity.
3. Improves availability of phosphorus and other nutrients.
4. Enhances microbial activity and nitrogen fixation.
5. Improves soil structure and root growth.

SECTION B

3. Discuss approaches that may be used to increase phosphorus availability in tropical soils (15 marks)

1. Application of phosphorus fertilizers (DAP, TSP, SSP).
2. Band placement near roots to reduce fixation.
3. Liming acidic soils to reduce Al and Fe fixation of P.
4. Use of rock phosphate in acidic soils.
5. Incorporation of organic matter to release organic acids.
6. Application of compost and manure.
7. Use of phosphate-solubilizing bacteria.
8. Inoculation with mycorrhizal fungi.
9. Crop rotation with legumes.
10. Intercropping systems that enhance P uptake.
11. Conservation tillage to maintain organic matter.
12. Balanced fertilization to avoid nutrient imbalances.
13. Use of P-efficient crop varieties.
14. Proper timing of P application at planting.
15. Soil testing for site-specific fertilizer recommendations.
16. Based on the image, I can see two exam questions about nitrogen fixation and soil fertility. Here are comprehensive answers:
- 17.

18. Question 4: Process Factors Limiting Biological Nitrogen Fixation in Smallholder Farms (15 marks)

19. **Answer:**

20. Biological nitrogen fixation (BNF) is the process by which atmospheric nitrogen is converted into ammonia by nitrogen-fixing microorganisms. In smallholder farms, several process factors limit this crucial function:
21. **1. Soil pH** Nitrogen-fixing bacteria (Rhizobium) and Azospirillum are sensitive to soil acidity. Most perform optimally at pH 6.0-7.0. Acidic soils (pH < 5.5) common in smallholder farms inhibit bacterial survival and nodule formation, reducing nitrogen fixation efficiency.
22. **2. Soil Moisture** BNF requires adequate soil water. Both drought stress and waterlogging limit oxygen availability and microbial activity. Smallholder farms with poor water management reduce nodulation and nitrogen fixation rates.

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23. **3. Soil Temperature** Nitrogen-fixing bacteria have optimal temperature ranges (25-30°C). In cooler climates or seasons, biological activity decreases significantly, slowing BNF processes.
24. **4. Soil Nutrients** Deficiencies in phosphorus, potassium, sulfur, and molybdenum limit BNF. Phosphorus is particularly critical for energy metabolism in nitrogen-fixation processes. Smallholder farms often lack fertilizer inputs.
25. **5. Oxygen Availability** While nitrogen fixers need anaerobic conditions in nodules, the nodule plant cells require oxygen. Poor soil aeration limits this balance, reducing overall fixation efficiency.
26. **6. Inoculation and Microbial Populations** Lack of effective Rhizobium strains and low microbial populations in uncultivated soils limit BNF. Smallholders may not have access to quality inoculants.
- 27.

28. Question 5: Organic Materials for Enhancing Soil Fertility in Smallholder Farms (15 marks)

29. **Answer:**

30. Smallholder farmers can utilize various organic materials to improve soil fertility sustainably:
31. **1. Crop Residues and Crop Biomass** Plant residues (maize stover, cereal stalks, legume haulms) improve organic matter, increase water retention, and add nutrients. They also suppress weeds and regulate soil temperature when incorporated or used as mulch.
32. **2. Compost** Well-decomposed organic matter from farm waste, kitchen scraps, and plant material improves soil structure, water-holding capacity, and provides slow-release nutrients. It increases microbial activity and nutrient availability.
33. **3. Green Manures and Cover Crops** Legume cover crops (Sesbania, Calliandra) fix atmospheric nitrogen and, when incorporated, increase soil nitrogen content. Non-legumes improve organic matter.
34. **4. Livestock Manures** Animal dung (cattle, poultry, goat) is rich in nitrogen, phosphorus, and potassium. It enhances soil structure, increases water retention, and improves microbial populations. Fresh or composted forms can be used.
35. **5. Mulches and Organic Mulching Materials** Agricultural residues and grass mulches conserve moisture, reduce temperature fluctuations, suppress weeds, and gradually decompose to increase organic matter and soil nutrients.
36. **6. Biosolids and Organic Waste Materials** Food processing waste, brewery waste, and agro-industrial byproducts contain nutrients and organic matter valuable for soil enrichment.
37. **7. Vermicompost** Worm-processed organic material is nutrient-rich and microbially active, improving soil structure and fertility with high nutrient availability.
38. **8. Tree Leaf Litter and Agroforestry Biomass** Fallen leaves from nitrogen-fixing trees improve soil organic matter and nutrient cycling in agroforestry systems.
39. **Benefits:** These materials sustainably improve soil structure, increase water-holding capacity, enhance nutrient availability, promote beneficial microbes, and reduce dependency on synthetic fertilizers in resource-limited smallholder farms.

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Exam Answers - Soil Fertility Questions

Question 2: Four Reasons Why Inorganic Inputs/Mineral Fertilizers Are Recommended in Less Fertile Soils (8 marks)

Answer:

1. Rapid Nutrient Availability Inorganic fertilizers provide nutrients in readily available forms that plants can quickly absorb. In less fertile soils where available nutrient levels are critically low, this immediate nutrient supply is essential to prevent crop failure and support plant growth quickly.

2. High Nutrient Concentration Mineral fertilizers contain high concentrations of macronutrients (N, P, K) and micronutrients, allowing farmers to apply precise quantities to meet crop demands. This is economical for small land areas and addresses severe nutrient deficiencies in less fertile soils.

3. Consistency and Predictability Inorganic fertilizers have standardized nutrient content and composition, allowing farmers to calculate exact nutrient requirements. This predictability is crucial in less fertile soils where nutrient status is uncertain.

4. Cost-Effectiveness for Nutrient Replenishment In severely depleted soils, the quantity of organic matter needed would be prohibitively expensive. Mineral fertilizers provide an economical solution to quickly restore soil fertility and improve crop productivity.

Question 3: Approaches to Increase Phosphorus Availability in Tropical Soils (15 marks)

Answer:

Tropical soils are typically highly weathered with low phosphorus availability due to fixation by iron and aluminum oxides. Several approaches address this:

1. Soil Acidification/pH Management Tropical soils are often neutral to alkaline. Lowering soil pH through sulfur application reduces P fixation by Al and Fe hydroxides, increasing soluble phosphorus. Acidification can be achieved through sulfur or acidifying fertilizers.

2. Application of Rock Phosphate Rock phosphate is most effective in acidic tropical soils where it dissolves slowly, providing long-term P availability. Combination with organic matter enhances dissolution rates.

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3. Use of Superphosphates Triple super phosphate (TSP) and single super phosphate (SSP) contain highly soluble phosphorus forms that overcome fixation problems, providing immediate availability in tropical soils.

4. Organic Matter Addition Compost, farmyard manure, and crop residues increase organic acids that compete with phosphate for fixation sites on Fe and Al oxides, reducing P fixation and improving availability. Organic matter also improves soil structure and water retention.

5. Biological Approaches - Mycorrhizal Associations Arbuscular mycorrhizal fungi (AMF) form symbiotic relationships with plant roots and access fixed phosphorus. Promoting AMF through reduced tillage and organic matter increases plant P availability.

6. Phosphate-Solubilizing Microorganisms Bacteria and fungi (Bacillus, Pseudomonas, Aspergillus) produce organic acids that dissolve fixed phosphorus. Inoculation or stimulation of these organisms improves P availability.

7. Integrated Nutrient Management Combining inorganic P fertilizers with organic matter and biological approaches optimizes availability. Co-application of nitrogen and potassium improves phosphorus uptake efficiency.

8. Liming (Selective Application) In very acidic tropical soils, mild liming can optimize pH for P availability (pH 6.0-7.0) without causing excessive fixation.

Question 4: Soil Organic Matter, pH Effects, and Organic Fertilizers (15 marks total)

a) How Cropping and Tillage Influence Soil Organic Matter (5 marks)

Answer:

Cropping Effects:

- **Continuous cropping** removes organic matter annually through crop harvest, reducing input back to soil and depleting organic matter stocks over time.
- **Legume crops** (pulses, beans) through biological nitrogen fixation add organic matter through residues and root biomass, increasing soil organic matter.
- **Mono-cropping** depletes specific nutrients and reduces organic matter diversity, while **intercropping and crop rotation** with legumes enhance organic matter accumulation.
- **Crop residue management** – leaving residues on soil increases organic matter, while removal reduces it.

Tillage Effects:

- **Conventional/intensive tillage** exposes organic matter to oxidation and microbial decomposition, rapidly decreasing soil organic matter content.

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- **Reduced/conservation tillage** minimizes soil disturbance, slowing organic matter decomposition and maintaining higher organic matter levels.
 - **No-tillage systems** preserve soil structure and maximize organic matter accumulation through minimal disturbance.
-

b) How Soil pH Influences Phosphorus Availability (5 marks)

Answer:

Soil pH critically affects phosphorus availability through chemical fixation mechanisms:

In Acidic Soils (pH < 6.0):

- Phosphorus is fixed by iron (Fe^{3+}) and aluminum (Al^{3+}) oxides/hydroxides
- Formation of insoluble Fe-P and Al-P compounds reduces plant-available phosphorus
- Excessive acidity reduces phosphorus availability despite adequate total P content

In Alkaline/Neutral Soils (pH 7.0-8.5):

- Phosphorus is fixed by calcium (Ca^{2+}) and magnesium (Mg^{2+})
- Forms insoluble Ca-P and Mg-P precipitates
- High pH reduces soluble phosphorus availability

Optimal pH Range (pH 6.0-7.0):

- Minimum P fixation by Al, Fe, and Ca
- Maximum phosphorus solubility and plant availability
- Lowest fixation by competing cations

pH Influence Mechanism:

- pH affects the charge on soil particles and phosphate ions
 - At optimal pH, electrostatic repulsion prevents P fixation
 - Changes in pH shift equilibrium between soluble and fixed phosphorus forms
-

c) Ways Organic Fertilizers Contribute to Plant Nutrient Availability (5 marks)

Answer:

1. Direct Nutrient Supply Organic fertilizers contain N, P, K and micronutrients. Decomposition releases these nutrients in available forms for plant uptake.

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2. Enhanced Microbial Activity Organic matter stimulates beneficial soil microorganisms that mineralize organic nutrients into plant-available inorganic forms (N, P, S).

3. Organic Acid Production Decomposing organic matter produces organic acids (citric, acetic) that:

- Dissolve fixed phosphorus and micronutrients
- Reduce aluminum and iron toxicity
- Chelate unavailable nutrients, making them available

4. Improved Soil pH Organic matter buffers soil pH, maintaining optimal conditions for nutrient availability and reducing fixation problems.

5. Enhanced Cation Exchange Capacity (CEC) Organic matter increases CEC, improving nutrient retention and reducing nutrient leaching losses, ensuring prolonged availability.

6. Improved Soil Structure and Water Retention Better soil structure increases root penetration and water availability, allowing roots better access to nutrients and improving nutrient uptake efficiency.

Question 5: Mineral Fertilizer Classification and Characteristics (15 marks)

a) Classification of Mineral Fertilizers by Number of Nutrients (3 marks)

Answer:

1. Straight/Single Fertilizers

- Contain only one primary nutrient (N, P, or K)
- Examples: Urea (46% N), Ammonium sulfate (21% N), Rock phosphate (18-30% P_2O_5)

2. Binary/Compound Fertilizers

- Contain two primary nutrients in combined form
- Examples: DAP (18-46-0, N-P-K), MAP (11-52-0)

3. Ternary/Complex Fertilizers

- Contain all three primary nutrients (N, P, K) in one granule
- Examples: NPK 15-15-15, NPK 10-10-10

4. Multi-nutrient/Specialty Fertilizers

- Contain primary nutrients plus micronutrients (Zn, Cu, B, Mn, Mo)

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- Examples: NPK + micronutrient blends
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b) Characteristics and Recommended Application Modes:

i. Rock Phosphate (3 marks)

Characteristics:

- Natural mineral ore containing 18-30% P_2O_5
- Insoluble/poorly soluble form
- Slow-release nutrient source
- Contains impurities (Ca, Si, Al)
- Lower cost than processed phosphates
- pH neutral to slightly alkaline
- Effective mainly in acidic soils where it dissolves

Recommended Application Mode:

- **Basal/incorporated application** – Mix thoroughly with soil during land preparation (6-8 weeks before planting) to allow time for dissolution
 - Apply in **acidic soils (pH < 6.5)** for better effectiveness
 - **Broadcasting and incorporation** preferred over localized placement
 - Suitable for **perennial crops** due to slow release nature
 - Apply **2-3 years before cultivation** in highly alkaline soils
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ii. Calcium Ammonium Nitrate (CAN) (3 marks)

Characteristics:

- Binary fertilizer: 26% N + calcium
- Highly soluble crystalline form
- Water-soluble, immediate nutrient availability
- Neutral pH (doesn't acidify soil)
- Reduces soil acidity (contains calcium)
- Hygroscopic (absorbs moisture) – requires proper storage
- More expensive than urea
- Suitable for high-value crops

Recommended Application Mode:

- **Top-dressing/split application** – Apply in 2-3 splits during growing season for optimal uptake
- **Foliar spray** – Dilute solution can be sprayed on foliage for emergency nitrogen replenishment
- **Broadcasting** on moist soil before rainfall

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- Apply **7-10 days before flowering** for reproductive tissue development
- Suitable for **field vegetables, fruits, tea, coffee**
- Keep away from moisture during storage

iii. Muriate of Potash (MOP) (3 marks)

Characteristics:

- Potassium chloride (KCl): 60% K₂O, 47% K
- Highly soluble salt form
- Contains chloride ion (problematic for sensitive crops)
- Rapid availability
- Pink/reddish granules
- Hygroscopic – requires dry storage
- Economical potassium source
- May increase soil salinity with repeated application

Recommended Application Mode:

- **Basal application** – Incorporate 3-4 weeks before planting as main potassium source
- **Avoid foliar spray** – Chloride causes leaf burn
- **Broadcasting and incorporation** method
- Apply in **well-drained soils** to prevent salt accumulation
- **Avoid sensitive crops** (tobacco, grapes, citrus) due to chloride toxicity
- Suitable for **cereals, pulses, oil crops, root crops**
- Split application in **2-3 doses** for crops with prolonged nutrient demand

iv. Triple Super Phosphate (TSP) (3 marks)

Characteristics:

- Processed phosphate fertilizer: 46% P₂O₅
- Highly soluble/available form
- Acidic pH (requires lime if soil already acidic)
- Rapid nutrient availability
- No calcium content (unlike SSP)
- More expensive than rock phosphate
- Concentrated form (high nutrient density)
- Hygroscopic – requires proper storage
- Effective across a range of soil pH values

Recommended Application Mode:

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- **Basal/localized placement** – Apply in bands or rows near root zone (planting holes) for concentrated nutrient availability
- **Broadcasting** in soils with pH > 7.0
- Apply **before planting** or at planting time for immediate availability
- **Combine with potassium** for complete nutrition (NPK fertilization)
- Excellent for **high-value crops** (vegetables, fruits, flowers)
- Can be used as **foliar feed** (diluted solution) for quick nutrient response
- Preferred in **neutral to alkaline soils** where rapid phosphorus availability is needed

Summary of Application Methods by Fertilizer Type:

Fertilizer	Primary Method	Secondary Method
Rock Phosphate	Incorporated basal (acidic soils)	Long-term placement
CAN	Top-dressing/split	Foliar spray
MOP	Basal incorporation	Split application
TSP	Basal/localized placement	Broadcasting (alkaline soils)

Question 1: Short Notes on Soil and Fertility Concepts (10 marks)

i. Basic Saturation (2 marks)

Answer:

Basic saturation is the percentage of the soil cation exchange capacity (CEC) occupied by basic cations (Ca^{2+} , Mg^{2+} , K^{+} , Na^{+}) expressed as a proportion of total CEC.

Formula: Basic Saturation (%) = (Sum of basic cations / CEC) $\times$ 100

Significance:

- Values >80% indicate fertile, neutral soils with good nutrient availability
- Values <50% indicate acidic, infertile soils with high exchangeable aluminum
- Directly affects soil pH and nutrient availability
- Used to assess soil fertility and lime requirement

ii. Incomplete Fertilizers (2 marks)

Answer:

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Incomplete fertilizers are mineral fertilizers that contain only **one or two of the primary macronutrients** (N, P, K), lacking the complete NPK profile.

Examples:

- Urea (N only) – 46% N
- Rock phosphate (P only) – 18-30% P_2O_5
- Muriate of potash (K only) – 60% K_2O
- DAP (N and P only) – 18-46-0

Disadvantages:

- Cannot provide complete nutrition alone
- Require combination with other fertilizers
- Less convenient for farmers
- Used as supplementary sources for specific deficiencies

iii. Critical Range of Nutrient Level (2 marks)

Answer:

The critical range (or critical level) is the **specific soil nutrient concentration range below which crop yield decreases significantly and above which additional nutrient application produces diminishing returns.**

Characteristics:

- Lies between **deficiency level** (plant symptoms appear) and **toxicity level** (excess causes damage)
- Varies by crop type, soil type, and environmental conditions
- Nutrient availability in this range ensures optimal plant growth and yield
- Used to make fertilizer recommendations

Example: Critical P level in soil may be 10-15 ppm; below this, deficiency symptoms appear; above 20 ppm, additional P shows minimal yield increase

iv. Disadvantage of Blended Fertilizers (2 marks)

Answer:

Blended fertilizers are mixtures of straight fertilizers (N, P, K) physically combined to create specific ratios.

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Main Disadvantages:

1. **Uneven Application and Segregation**
 - Different particle sizes cause segregation during handling and transport
 - Lighter particles (like urea) separate from heavier ones, causing non-uniform distribution to field
 - Results in uneven nutrient supply to plants
2. **Poor Nutrient Ratio Consistency**
 - Cannot guarantee exact nutrient composition in every application
 - Difficult to achieve precise NPK ratios desired by farmers
3. **Higher Cost**
 - More expensive than complex/compound fertilizers
 - Requires manual or mechanical blending (labor-intensive)
4. **Storage and Handling Issues**
 - Blends are hygroscopic and absorb moisture during storage
 - Caking and hardening occur, making application difficult

v. Synthetic Nitrogen Fixation with Legumes (2 marks)

Answer:

This question appears to ask about "synthetic" interaction with nitrogen fixation, likely referring to **symbiotic nitrogen fixation** (relationship between legumes and Rhizobium bacteria).

Symbiotic Nitrogen Fixation Process:

- **Legume plants** (beans, peas, clover) form **root nodules** housing Rhizobium bacteria
- Bacteria convert atmospheric N_2 to ammonia (NH_3) in anaerobic nodule environment
- Legume provides carbohydrates from photosynthesis; bacteria provide fixed nitrogen
- Plant absorbs ammonia; excess returned to soil when nodules decompose
- Fixation rates: 100-300 kg N/ha/year depending on legume type and conditions

Agricultural Significance:

- Reduces nitrogen fertilizer requirement
- Improves soil fertility through nitrogen enrichment
- Used in crop rotation and intercropping systems

Question 2: Three Indicators of Soil Fertility Status in a Farmer's Field (3 marks)

Answer:

1. Visual/Physical Indicators:

- **Soil Color** – Dark-colored soils indicate high organic matter; pale/light soils suggest low fertility

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- **Soil Structure** – Crumbly, well-aggregated soil indicates good fertility and organic matter; compacted, hard soils indicate poor fertility
- **Soil Texture Feel** – Loamy soils (friable) are more fertile than sandy (loose) or clay (sticky) soils
- **Presence of Organisms** – Abundant earthworms, termites, and visible root activity indicate fertile soils

2. Plant/Crop Performance Indicators:

- **Crop Growth Rate** – Vigorous, healthy crop growth indicates adequate soil nutrients
- **Plant Color** – Dark green foliage suggests good nitrogen availability; yellowing suggests deficiencies
- **Weed Abundance and Type** – Presence of nutrient-demanding weeds indicates fertility; nitrogen-loving weeds (Amaranthus) indicate nitrogen availability
- **Crop Yield** – Historical yield data shows long-term fertility trends

3. Soil Chemical Indicators (Laboratory/Field Tests):

- **Soil Organic Matter Content** – >3% indicates fertile soil
- **Soil pH** – pH 6.0-7.0 indicates optimal nutrient availability
- **Available Nutrients** – High levels of exchangeable Ca, Mg, K and available P indicate fertility
- **CEC and Base Saturation** – High CEC (>15 cmol/kg) and base saturation >80% indicate fertile soils
- **NPK Content** – Total nitrogen >0.2%, available phosphorus >10 ppm, potassium >100 ppm indicate fertility

Question 3: Deficiency Symptoms of Phosphorus in Plants (3 marks)

Answer:

Phosphorus deficiency affects energy metabolism and structural development. Symptoms include:

Early/General Symptoms:

- **Stunted Growth** – Reduced plant height and overall biomass development
- **Purple/Red Coloration** – Characteristic reddish or purple discoloration on leaves, stems, and petioles due to anthocyanin accumulation
- **Dark Green Foliage** – Leaves appear abnormally dark green, often small and thick

Leaf Symptoms:

- **Small Leaves** – Reduced leaf size and delayed leaf expansion
- **Leaf Scorching** – Necrotic spots on leaf tips and margins, particularly on older leaves
- **Leaf Curling** – Downward curling of leaf margins
- **Premature Leaf Fall** – Early senescence and defoliation

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Root and Structural Symptoms:

- **Poor Root Development** – Reduced root branching and shallow root system
- **Delayed Flowering/Maturity** – Delayed reproductive development
- **Weak Stems** – Thin, brittle stems susceptible to lodging
- **Poor Fruit/Seed Development** – Reduced fruit set, seed number, and seed quality

Specific Crop Symptoms:

- **Cereals:** Purple/red leaves, reduced tillering
- **Legumes:** Yellow-green leaves with purple veining
- **Root Crops:** Small roots with poor storage

Question 4: Three Harmful Effects of Excessive Nitrogen Supply on Plants (3 marks)

Answer:

Excessive nitrogen application produces several detrimental effects:

1. Excessive Vegetative Growth (Luxury Consumption)

- **Excessive leaf and stem growth** at expense of reproductive development
- **Delayed flowering and fruiting** – Extended vegetative phase reduces economic yield
- **Poor seed/grain quality** – Excessive protein production with poor harvest-index
- Results in **poor yield quality** despite apparent vigorous growth

2. Disease and Pest Susceptibility

- **Increased disease incidence** – Excessive nitrogen promotes soft, succulent tissues attractive to fungal pathogens (powdery mildew, leaf spots)
- **Enhanced pest attack** – Insects prefer nitrogen-rich plants; susceptibility to aphids, armyworms increases
- **Reduced plant immunity** – Weakened disease resistance mechanisms due to imbalanced nutrients
- **Increased lodging damage** – Soft tissues and weak stems susceptible to physical damage

3. Reduced Nutrient Uptake and Imbalance

- **Phosphorus and potassium deficiency** – Excessive N reduces P and K absorption and translocation
- **Micronutrient deficiency** – High N reduces Zn, Cu, Mn availability and uptake
- **Nutrient Imbalance** – Creates unfavorable N:P:K ratios affecting plant metabolism
- **Yield reduction** – Despite high N, overall growth is limited by deficiency of other nutrients

Additional Harmful Effects:

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- **Environmental Pollution** – Nitrate leaching contaminates groundwater
- **Salinity Stress** – High nitrogen increases soil salt concentration, causing osmotic stress
- **Poor Fruit Quality** – Reduced sugar content, flavor, and shelf-life in fruits and vegetables

Question 5: State Four (3 Ways) Through Which Soil is Lost from Plant Nutrients (3 marks)

Answer:

Soil loses plant nutrients through several mechanisms (pathways):

1. Leaching/Percolation

- **Water movement** through soil carries soluble nutrients (NO_3^- , K^+ , SO_4^{2-}) below root zone
- **Occurs in sandy soils** with low CEC and high rainfall
- **Nitrate leaching** is most significant, particularly for nitrogen
- Causes loss of mobile nutrients even when abundantly applied
- **Prevention:** Use slow-release fertilizers, maintain organic matter for nutrient retention

2. Crop Removal/Offtake

- **Harvesting crops** removes substantial quantities of nutrients accumulated in plant biomass
- **Grain crops** remove significant N, P, K in seeds/grain
- **Removal rates** vary: cereal grain (50-100 kg N, 15-30 kg P, 30-60 kg K per ton)
- **Crop residue removal** (burning, collection for fodder/fuel) prevents nutrient return to soil
- **Root removal** in horticultural crops causes significant nutrient loss

3. Erosion (Water and Wind Erosion)

- **Soil loss** carries nutrients-rich topsoil layer away from field
- **Water erosion** caused by rainfall runoff on sloping lands removes organic matter and clay-bound nutrients
- **Wind erosion** particularly in arid regions removes fine nutrient-rich particles
- **Nutrient loss** proportional to soil loss; organic matter loss is particularly significant
- **Prevention:** Contour cultivation, terracing, windbreaks, mulching

4. Gaseous Loss (Volatilization and Denitrification)

- **Ammonia volatilization** – NH_3 gas escapes from urea, ammonium sulfate, farmyard manure (especially on calcareous soils)
- **Nitrogen gas loss** – Denitrification reduces nitrate to $\text{N}_2/\text{N}_2\text{O}$ in anaerobic conditions (waterlogged soils)
- **Losses of 10-70%** of applied nitrogen possible under unfavorable conditions
- **Sulfur loss** – H_2S volatilization in anaerobic conditions
- **Prevention:** Incorporate nitrogen sources quickly, improve drainage, use nitrification inhibitors

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Additional Pathway (if asked for more):

5. Microbial Immobilization

- **Microorganisms** incorporate nutrients into biomass, making them unavailable temporarily
- **High C:N ratios** in crop residues cause nitrogen immobilization
- While nutrients are eventually released, temporary unavailability occurs

Summary Table: Nutrient Loss Mechanisms

Mechanism	Nutrients Affected	Conditions	Prevention
Leaching	N, K, S, micronutrients	High rainfall, sandy soils, low CEC	Slow-release fertilizers, increase OM
Crop Removal	N, P, K, all nutrients	Harvesting, residue removal	Retain residues, crop rotation
Erosion	All, especially OM	Sloping land, high rainfall, wind	Contour cultivation, terracing, mulch
Volatilization	N (as NH ₃)	Calcareous soils, high pH	Incorporate quickly, use urea formulations
Denitrification	N (as N ₂ /N ₂ O)	Waterlogged, anaerobic soils	Improve drainage, proper timing
